Calculation of the difference between the areas illuminated by the first and last excitation flashlet of a single LIFT measurement

The distance *d* travelled by the robot during time interval Δt at a forward speed of *v* is given by the formula (1):

$$d = v \cdot \Delta t \tag{1}$$

From the general equation of a circle with center (m_1, m_2) and radius *r* (2),

$$(x - m_1)^2 + (y - m_2)^2 = r^2 \tag{2}$$

we can derive the formula for the part of the area illuminated by the 300th flashlet (i.e. the last excitation flashlet) that was *not* illuminated by the 1st (and all subsequent) flashlets associated with the same F_q'/F_m' measurement:

Let (- *d*/2, 0) be the center of the excitation beam at *t* = 0 and (*d*/2, 0) the center of the excitation beam at *t* = 750 µs so that the circles delimiting the outer boundary of the excitation beam intersect on the *y* axis (i.e. at *x* = 0).

Then the equation of the points on the boundary line of the excitation beam at *t* = 0 µs is given by (3)

$$(x + \tfrac{d}{2})^2 + (y - 0)^2 = r^2 \tag{3}$$

Solving (3) for *y* yields

$$y = \sqrt{r^2 - \left(x + \frac{d}{2}\right)^2} \tag{4}$$

Analogously, the equation of the points on the boundary line of the excitation beam at *t* = 750 µs is given by (5)

$$(x - \tfrac{d}{2})^2 + (y - 0)^2 = r^2 \tag{5}$$

Solving (5) for *y* yields

$$y = \sqrt{r^2 - \left(x - \frac{d}{2}\right)^2} \tag{6}$$

The area in between the two circle boundary lines on the right side of the y axis (*A'*) can then be calculated as

$$A' = 2 \cdot \left(\int_0^{r+\frac{d}{2}} \sqrt{r^2 - \left(x - \frac{d}{2}\right)^2}\, dx - \int_0^{r-\frac{d}{2}} \sqrt{r^2 - \left(x + \frac{d}{2}\right)^2}\, dx \right) \tag{7}$$

Using a calculator with a computer algebra system, (7) can be simplified to

$$A' = 2 \cdot r^2 \cdot arcsin_{RAD}\left(\frac{d}{2 \cdot r}\right) + \frac{d}{2} \cdot \sqrt{4 \cdot r^2 - d^2} \tag{8}$$

Substituting (1) into (8) yields

$$A' = 2 \cdot r^2 \cdot arcsin_{RAD}\left(\frac{v \cdot \Delta t}{2 \cdot r}\right) + \frac{v \cdot \Delta t}{2} \cdot \sqrt{4 \cdot r^2 - (v \cdot \Delta t)^2} \tag{9}$$

The total area illuminated by the excitation beam at one moment is given by (10)

$$A = r^2 \cdot \pi \tag{10}$$

Meaning that the proportion of the area illuminated by the last excitation flashlet that was *not* already illuminated by the first excitation flashlet can be expressed as (11)

$$\frac{A\prime}{A} = \frac{2}{\pi} \cdot arcsin_{RAD}\left(\frac{v \cdot \Delta t}{2 \cdot r}\right) + \frac{v \cdot \Delta t}{2 \cdot r^2 \cdot \pi} \cdot \sqrt{4 \cdot r^2 - (v \cdot \Delta t)^2} \quad (11)$$

Substituting 0.5 ms^{-1} for *v*, 0.00075 s for Δt and 0.01 m for *r* yields

$$\frac{A\prime}{A} \approx 0.024 \quad (12)$$

This means that the fluorescence yield of the last excitation flashlet is measured without any additional error due to the forward motion of the robot on approx. 97.6% of *A* at v = 0.5 ms^{-1}. But even on the remaining approx. 2.4% of A (i.e. *A'*), the measurement error on the fluorescence yield of the last excitation flashlet will mostly be numerically small due to the fast initial increase in fluorescence yield per flashlet during the excitation phase of the LIFT measurement (see Figure 1B in our manuscript).